

YASH BAVADIYA

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EDUCATION

Indian Institute of Technology, Jodhpur

Aug 2024 – Jul 2028

Bachelor's Degree - Computer Science and Engineering - CGPA - 7.21

Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, SQL

Backend & Distributed Systems: Node.js, Express, REST APIs, WebSockets, Redis (pub/sub, BullMQ), Message Queues, PostgreSQL, MySQL, MongoDB, Prisma, Docker, Nginx, Linux/UNIX, TCP/IP

AI & Data: LLM Prompting & Evaluation, Retrieval-Augmented Generation, FAISS Vector Search, Hugging Face Transformers, Information Extraction, Regex/NLP, PyTorch, Pandas

Tools: Git, GitHub Actions, Postman, Vercel, React, Next.js, Qt (C++), VS Code

Core Competencies: Data Structures and Algorithms, Distributed Systems, System Design, Operating Systems, Debugging, Technical Writing (RFCs, PRs, incident notes)

EXPERIENCE

Google Summer of Code 2026

May 2026 – Jun 2026

Kdenlive @KDE Community | Open Source Contributor, C++ and Qt

Remote

- Secured selection for **GSoC 2026** among 1,200 global contributors to implement Kdenlive's **RGB curve editor, gradient editor, and speed ramp widgets** in C++ and Qt.
- Integrated **20 contributions** across Kdenlive and upstream repositories, resolving an **11-year-old bug** via a gap-detection algorithm shipping to **41,000 monthly Flathub installs**.

Undergraduate Researcher

Jul 2025 – Nov 2025

Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder

IIT Jodhpur

- Engineered an offline **Retrieval-Augmented Generation pipeline** in **Python**, integrating FAISS vector search with 4-bit quantized Phi-3 and Llama-3.1 for document information extraction.
- Built an **LLM evaluation harness** benchmarking prompt and retrieval strategies across **8 ROUGE metrics** (+71.77% over baseline), reaching **3 second query latency** on 300-page PDFs for **15 users**.

PROJECTS

Flock | [GitHub](#) | [npm](#) | TypeScript, WebSocket, Redis, React, Docker

Feb 2026 – Jun 2026

- Released 3 npm packages enabling multiplayer cursors and live presence in web applications via a **10-line integration**, installable as a **55MB** self-hosted Docker image or npx.
- Implemented a horizontally scalable **Redis pub/sub message relay** across server instances with keyspace-notification-driven eviction, reducing stale presence state to under **30 seconds** on ungraceful disconnects.
- Delivered cursor position smoothing with continuous mid-animation target updates, exponential backoff reconnection, and **58 tests** covering throttling, time-to-live eviction, and message delivery.

DaemonDoc | [GitHub](#) | [Live](#) | TypeScript, Express, MongoDB, Redis (BullMQ), Docker

Jan 2026 – May 2026

- Shipped an **LLM-driven documentation agent** triggered by GitHub webhooks, sustaining an **85% success rate** across **959 production runs** over 68 repositories for **50 users**.
- Architected a **Redis (BullMQ) queue-backed job pipeline** with exponential backoff, retries, and fault isolation, cutting **LLM token costs by 70%** via source filtering during prompt construction.
- Orchestrated **4 services** with Docker Compose, configuring an Nginx reverse proxy and GitHub Actions pipeline to enable zero-downtime deployments.

IIT Jodhpur PHC System | [GitHub](#) | Node.js, Express, PostgreSQL, Prisma, Docker

Jan 2026 – Apr 2026

- Spearheaded backend architecture replacing a paper workflow with a role-based platform of **63 REST endpoints** over a **20-model relational schema** (PostgreSQL/Prisma), with JWT and LDAP authentication across **6 roles**.
- Verified reliability via **48 test cases** and load tests at **20 concurrent users**, securing production deployment approval projecting **5 staff-hours saved per day**.

COURSEWORK

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|----------------------------------|-------------------------------|------------------------------|
| • Data Structures and Algorithms | • Machine Learning | • Pattern Recognition |
| • Operating Systems | • Software Engineering | • Probability and Statistics |
| • Database Management Systems | • Object-Oriented Programming | • Discrete Mathematics |

ACHIEVEMENTS

- JEE Advanced 2024: AIR **3449**; JEE Main 2024: **99.65** percentile, AIR **5680**.
- Achieved Max Rating of **1429** on Codeforces; solved **450+** algorithmic problems on LeetCode in C++.
- Merged **84+ open-source contributions** across GitHub and GitLab, authoring issue reports, design notes, and reviewed pull requests for **Kdenlive**, **Vitest**, **Wagtail**, and **crocc** (Go).

LEADERSHIP

- Junior Core Team Member** at DevUp Labs, IIT Jodhpur; mentored 30 students and facilitated 2 open-source workshops.
- Core Member** at E-Cell, IIT Jodhpur; supported 3 entrepreneurship events and initiatives over 9 months.
- Core Member** at Programming Club, IIT Jodhpur; co-organized 5 contests and 10 sessions on competitive programming.